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Lecture Class-IV V Atomic and Molecular Simulations (CHE F497)

11th & 13th Aug 2026

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Module-II Quantum Chemical Methods

Born-Oppenheimer Approximation, Hydrogen and Multielectron Atoms, Hartree-Fock Theory and Basis Sets, Density Functional Theory, Geometry Optimization, Electronic Structure of Atoms and Molecules, Ab Initio MD, Application,

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Why Quantum Mechanics ?



- Simplest answer: we're living in a quantum world
- To make and control the Electronic, opto-electronic, optical devices
- To understand band gap, device physics
- Quantum mechanics is the basis for understanding of physical phenomena on atomic scale
- Numerous applications: semiconductor devices, laser, quantum optics
- Classical mechanics is unable to explain quantization
- The failure of classical physics to explain several microscopic phenomena—such as *blackbody radiation*, the *photoelectric effect*, *atomic stability*, and *atomic spectroscopy*—had cleared the way for seeking new ideas outside its purview.
- Quantum mechanics to understand the microscopic properties of atoms
- Light waves are quantized into particles called photons
- Color of visible light is associated with different wavelenaths

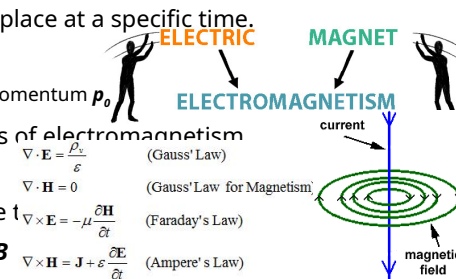
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Basics of Quantum Mechanics



Matter-Radiation Theory:

- Before the advent of Quantum mechanics, classical physics studied the Universe as a system consisting **matter** and **radiation**
- The matter are particles whose motion in time and space is governed by laws of mechanics formulated by Newton.
- In classical mechanics, objects exist in a specific place at a specific time.
 - At a given instant time t_0 ,
 - the state of classical system $\rightarrow$ position vector $\mathbf{r}_0$ and momentum $\mathbf{p}_0$
- On the other hand, radiation is governed by laws of **electromagnetism** formulated by **James Clark Maxwell**.
- The dynamical variables of the radiation field are $\mathbf{E}$ and $\mathbf{B}$



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Brief History:

Planck introduced concept of absorption and emission of radiation in discrete quanta

• 1900 (Planck):

- Explained Spectral Distribution of black-body radiation
- He then postulated that the energy of the radiation (of frequency ν) emitted by the oscillating charges (from the walls of the cavity) must come *only* in *integer multiples* of $h\nu$:

$$E = nh\nu, \quad n = 0, 1, 2, 3, \dots,$$

• He argued that the energy exchange between an electromagnetic wave of frequency ν and matter occurs only in integer multiples of $h\nu$, which he called the energy of a quantum, where h is a fundamental constant called Planck's constant.

Known as **Planck's quantization rule for energy or Planck's postulate.**

$$E = h\nu$$

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Brief History

Quantum Nature of Light

• 1900 (Planck):

- Explained *black-body radiation*
- Quantization of light: Max Planck proposed that light with frequency ν is emitted in *discrete* or *quantized* lumps of energy
- Assumed emission and absorption of discrete energy quanta of electromagnetic radiation
- He argued that the energy exchange between an electromagnetic wave of frequency ν and matter occurs only in integer multiples of $h\nu$, which he called the energy of a quantum, where h is a fundamental constant called Planck's constant.

$$E = h\nu$$

• 1905 (Einstein):

- Einstein recognized that Planck's idea of the quantization of the electromagnetic waves must be valid for light as well to understand the **photoelectric effect**
- Albert Einstein stated that the quantization was in fact inherent to the light, and that the lumps can be interpreted as particles, which we now call "photons."
- This proposal was a result of his work on the **photoelectric effect**, which deals with the absorption of light and the emission of electrons from a material.

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Basics of Quantum Mechanics

- Unlike matter, the Radiation field consists of waves.
- It was firmly believed (until the end of the nineteenth century and the beginning of the twentieth century) that all phenomena in physics could be explained in the general framework of matter-radiation theory.
- Atomic and subatomic structures of matter were explored which suggests non-classical behaviour of matter at the microscopic level.
- Quantum mechanics is a subfield of physics that describes the behavior of matters (atoms, electrons, photons) and energy in the scale of atomic and subatomic particles or waves
- In quantum mechanics, objects instead exist in a haze of probability; they have a certain chance of being at point A, another chance of being at point B and so on.
- In quantum mechanics, particles have wavelike properties, and a particular wave equation, the Schrodinger equation, governs how these waves behave.



image of quantum probability

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Particle Theory of Light

$$E = mc^2$$

Energy mass speed of light

- Blackbody radiation
- The photoelectric effect
- The Compton scattering

According to Wave Nature

frequency $\rightarrow v = \frac{c}{\lambda}$ $\leftarrow$ wavelength

speed of light

Wavelength and frequency are inversely related to each other

$$\text{Speed of light } \rightarrow c = \lambda \nu \leftarrow \text{Frequency}$$

($3 \times 10^8 \text{ m/s}$)

Wavelength

As wavelength increases, frequency decreases
As wavelength decreases, frequency increases

Dual nature of light

- Now, we accept that light has a dual nature.
- In some cases, light behaves like a wave, and in others, light behaves like a particle.

• **Wave Nature:** Diffraction and Interference

• **Particle Nature:** exhibits photoelectric effect & Compton effect

According to Quantum Nature

Energy E of one quantum of light (photon)

$$E = h\nu$$

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$$E = h\nu$$

$$E = mc^2$$

Energy mass speed of light

Combining quantum aspect to wave aspect: →

$$c = \lambda\nu$$

Momentum?

$$p = \frac{h\nu}{c}$$

The de Broglie Hypothesis

In 1924, de Broglie suggested that if waves of wavelength λ were associated with particles of momentum $p = h/\lambda$, then it should also work the other way round.....

A particle of mass m , moving with velocity v has momentum p given by:

$$p = mv = \frac{h}{\lambda}$$

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- microscopic particles, such as electrons, display wave behaviour
- What about macroscopic objects?
- Do they also display wave features? They surely do.

de Broglie wavelength

$$\lambda = \frac{h}{p}$$

macroscopic objects have extremely small wavelengths

The general rule is: whenever the de Broglie wavelength of an object is in the range of, or exceeds, its size, the wave nature of the object is detectable and hence cannot be neglected. But if its de Broglie wavelength is much too small compared to its size, the wave behavior of this object is undetectable.

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Wave-Particle Duality: Complementarity

Particle or Wave?

- electron and photons sometimes appear to behave as particles and sometimes as waves
- They appears to be a complementarity in the way one treats their behavior.
- Wave-particle duality is the concept in quantum mechanics that every particle or quantum entity may be described as either a particle or a wave.
- Neutrons, protons and electrons can seem like particles with a mass and momentum
- However, if one looks on an appropriate length or time-scale, they might exhibit the key characteristics of waves e.g., superposition and interference
- If we measure the particle properties of a quantum system, this will destroy its wave features, and vice versa.
- We can get either the wave property or the particle but not both of them together

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Quantum Mechanics: Matrix Mechanics

Historically, there were two independent formulations of quantum mechanics. The first formulation, called matrix mechanics, was developed by Heisenberg (1925) to describe atomic structure starting from the observed spectral lines. Inspired by Planck's quantization of waves and by Bohr's model of the hydrogen atom, Heisenberg founded his theory on the notion that the only allowed values of energy exchange between microphysical systems are those that are discrete: quanta. Expressing dynamical quantities such as energy, position, momentum and angular momentum in terms of matrices, he obtained an eigenvalue problem that describes the dynamics of microscopic systems; the diagonalization of the Hamiltonian matrix yields the energy spectrum and the state vectors of the system. Matrix mechanics was very successful in accounting for the discrete quanta of light emitted and absorbed by atoms.

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The second formulation, called wave mechanics, was due to Schrödinger (1926); it is a generalization of the de Broglie postulate. This method, more intuitive than matrix mechanics, describes the dynamics of microscopic matter by means of a wave equation, called the Schrödinger equation, instead of the matrix eigenvalue problem of Heisenberg. Schrödinger obtained a differential equation. The solutions of this equation yield the energy spectrum and the wave function of the system under consideration. In 1927 Max Born proposed his probabilistic interpretation of wave mechanics: he took the square moduli of the wave functions that are solutions to the Schrödinger equation and he interpreted them as probability densities.

- **A wave function (Ψ)** is a mathematical function that relates the location of an electron at a given point in space (identified by x , y , and z coordinates) to the amplitude of its wave, which corresponds to its energy.
- Thus each wave function is associated with a **particular energy E** .

- **Wave-particle duality** is deeply embedded into the foundations of **quantum mechanics**.
- In the formalism of the theory, all the information about a particle is encoded in its **wave function**, a complex-valued function roughly analogous to the amplitude of a wave at each point in space.
- This function evolves according to **Schrödinger equation**.

Wave Function

- **A wave function (Ψ) is a mathematical function that relates the location of an electron at a given point in space (identified by x , y , and z coordinates) to the amplitude of its wave, which corresponds to its energy.**
- The **state** of a **quantum system** at a given instant of time, t , is completely defined by a **function**, wave function of the particle
 - $\Psi(r, t)$ where r is the position vector $r = \{x, y, z\}$ and t is the time.
- **Statistical Interpretation:**
 - the wave function $\Psi(r, t)$ doesn't have any physical meaning of its own
 - **Probability (P)** amplitude at a given instant time, t , : the particle will be found in an infinitesimal volume element, $\Delta V = dx dy dz$

$$P = \int_V |\Psi(r, t)|^2 dV$$
 - This statistical interpretation of the wave function proposed by Max Born
 - $|\Psi(r, t)|^2 \rightarrow$ probability density for locating the particles in space at a given instant of time.

In 1927 Born interpreted $|\psi|^2$ as the *probability density* and $|\psi(\vec{r}, t)|^2 d^3r$ as the probability, $dP(\vec{r}, t)$, of finding a particle at time t in the volume element d^3r located between $\vec{r}$ and $\vec{r} + d\vec{r}$:

$$|\psi(\vec{r}, t)|^2 d^3r = dP(\vec{r}, t),$$

The Born-Oppenheimer approximation

- The electrons are much lighter than the nuclei ($m_e/m_H \approx 1/1836$) • their motion is much faster than the vibrational and rotational motions of the nuclei within the molecule.

→ The Schrödinger equation can then be divided into two equations:

- 1) One describes the *motion of the nuclei*.
- 2) The other one describes the *motion of the electrons* around the nuclei whose positions are fixed.

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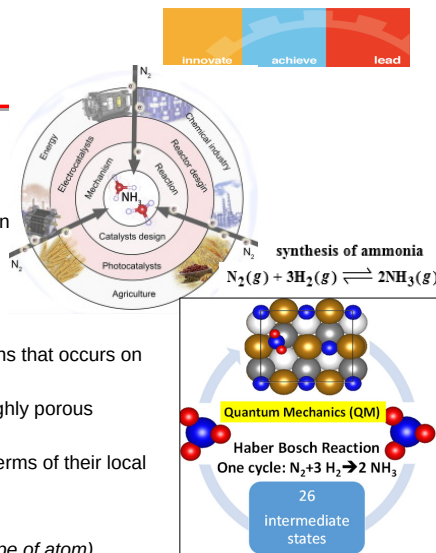
- In summary, **quantum mechanics** is the theory that can describe the dynamics of matter at the microscopic scale.
- It is vital for understanding the physics of solids, lasers, semiconductor and superconductor devices, plasmas, etc.
- In short, quantum mechanics is the founding basis of all modern physics: *solid state, molecular, atomic, nuclear, and particle physics, optics, thermodynamics, statistical mechanics, and so on.*
- Not only that, it is also considered to be the foundation of chemistry and biology.

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DFT: Applications

Ammonia Synthesis by Heterogeneous Catalysts

- Ammonia → fertilizer for agriculture
- > 100 million tons of NH_3 are produced commercially each year
- 1% of all energy used in the world is consumed in NH_3 production
- Temperature > 400 °C and Pressure > 100 atm
- Catalysts: metal e.g., iron (Fe) or ruthenium (Ru)
- These metal catalysts were identified 100 years ago
- But, much is still not known about the mechanism of the reactions that occurs on the surfaces of the catalysts
- To increase the surface area, metal NP dispersed throughout highly porous materials
- To understand the reactivity, characterize the surface atoms in terms of their local coordinates
- As difference in coordinates can cause difference in reactivity
- Overall surface reactivity = $f(\text{shape of NP \& reactivity of each type of atom})$

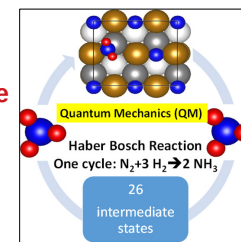


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DFT: Applications

Ammonia Synthesis by Heterogeneous Catalysts

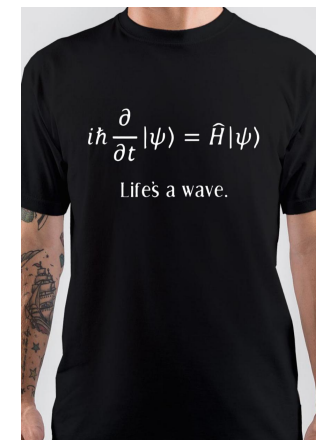
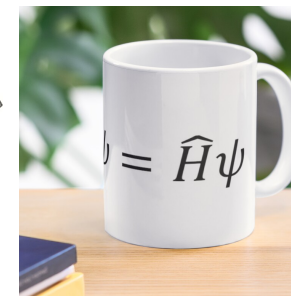
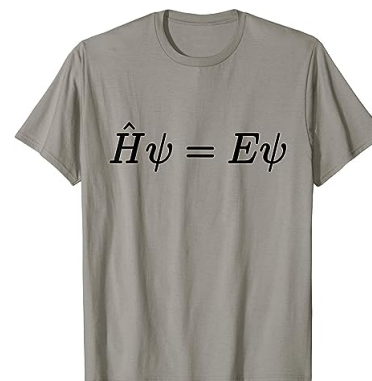
- Overall surface reactivity = $f(\text{shape of NP \& reactivity of each type of atom})$
- The surface of nanoparticles typically include atoms of various types (based on coordination)
- Can a direct connection be made between the shape and size of a metal nanoparticle and its activity as a catalysts for ammonia synthesis?**
- If yes, they can potentially lead to synthesis of improved catalysts
- DFT calculations** can give us most detailed answers to the question
- 12-26 distinct steps involve
- Rates of these steps strongly depend on the local coordination of the metal atoms that are involved
- Stability of many different local coordinates of surface atoms can be studied
- Which will allow to predict a detailed shape of the NPs as a function of particle size



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- **1998, Nobel Prize** in Chemistry → jointly to **Walter Kohn** for developing the foundations of **DFT** and **John Pople** for developing a quantum chemistry computer code for calculating the **electronic structure** of atoms and molecules
- **first** time that a Nobel prize in chemistry or physics was awarded for the development of a **numerical** method

This equation is in a nice form for putting on a T-shirt or a coffee mug



The Schrödinger Wave Equation

- The time-dependent Schrödinger wave equation which predicts the position of electron waves of one particular atom, moving in an electric field, is as follows :

$$\hat{H}\Psi(r_i, t) = i\hbar \frac{d}{dt} \Psi(r_i, t)$$

where,

or

$$\hat{H}\Psi = E\Psi$$

- Here the energy Hamiltonian ($\hat{H}$) is the function of Kinet energy (K.E) and potential energy (V), whereas the potential energy term includes electron-electron interaction and external potential energy

- Due to severe complexity, approximations are needed to solve the many-body interacting Schrödinger wave equation

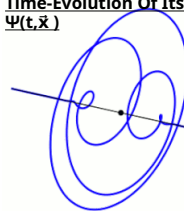
- $\hat{H}$ total energy Hamiltonian operator,
- Ψ is the wave function,
- i is the imaginary unit,
- $\hbar$ is the Planck's constant,
- E is the eigenvalue which defines energy of Ψ state,
- r is the position vector,
- t defines time.

DFT:

Time-Independent, Nonrelativistic Schrödinger Equation

$$\hat{H}\Psi = E\Psi$$

Time-Evolution Of Its Associated Wave-Function $\Psi(t, \vec{x})$



Complex plot of a **wave function** that satisfies the nonrelativistic Schrödinger equation with $V = 0$. In other words, this **corresponds to a particle traveling freely through empty space**

- In this equation, **H is the Hamiltonian Operator**
- Ψ is a **set of solutions**, or **eigenstates** of the Hamiltonian
- **Each** of these solutions, Ψ_n , has an associated **eigenvalue, E_n** , a real number that satisfies the eigenvalue equation.
- There are several well-known examples like the particle in a box or a harmonic oscillator where the **Hamiltonian** has a **simple** form
- And the **Schrödinger equation** can be solved exactly
- Where multiple electrons are interacting (the situation we are interested) with multiple nuclei is more complicated
- Complete description of the Schrödinger equation is used

DFT Calculations



- In this equation, H is the Hamiltonian Operator
- Ψ is a set of solutions, or eigenstates of the Hamiltonian

$$\hat{H}\Psi = E\Psi$$

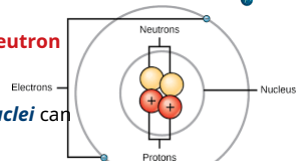
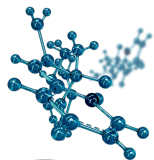
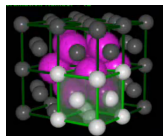
- Each of these solutions, Ψ_n has an associated eigenvalue, E_n , a real number, that satisfies the eigenvalue equation
- The detailed definition of the Hamiltonian depends on the physical system being described by the Schrödinger equation
- There are several well-known examples like the particle in a box or a harmonic oscillator where the Hamiltonian has a simple form
- And the Schrödinger equation can be solved exactly
- Where multiple electrons are interacting (the situation we are interested) with multiple nuclei is more complicated
- Complete description of the Schrödinger equation is used

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DFT Calculations



- Let us imagine, a situation → we would like to describe the properties of some well-defined collection of atoms (think *isolated molecule* or the *atoms* defining the *crystal* of an interesting *mineral*)
- One of the **fundamental things** we would like to know about these atoms is their **Energy** and more importantly how their **energy changes** if we **move** the **atoms** around
- To define the where an atom is → 1) we need to define where its **nucleus** is
 - 2) where the atom's **electrons** are
- Atomic nuclei are much heavier than individual electrons : → each **proton** or **neutron** has more than **1800 times** the mass of an **electron**
- **Electrons** respond much more **rapidly** to change in their surroundings than **nuclei** can
- So, we can split our physical question into two pieces

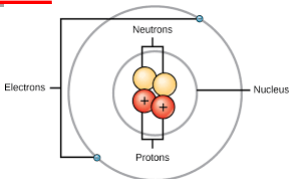


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DFT Calculations



- So, we can split our physical question into **two** pieces
- **1)** we solve for fixed positions of the **atomic nuclei**, 2) the equations that describe the **electron motion**
- For a given set of electrons moving in the field of a set of nuclei, we find the **lowest energy configuration**, or **state**, of the electrons
- The lowest energy state is known as the **ground state** of the electrons



- **Born-Oppenheimer Approximation** : → the separation of the nuclei and electrons into separate mathematical problems

- 1) One describes the **motion of the nuclei**.
- 2) The other one describes the **motion of the electrons** around the nuclei whose positions are fixed.

- If we have M nuclei positions ($R_1, \dots, R_M$), then we can express the ground-state energy, E , as a function of the positions of these nuclei, $E(R_1, \dots, R_M)$
- This function is known as the **adiabatic potential energy surface** of the atoms

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DFT Calculations:



» Schrödinger equation

$$\left[-\frac{\hbar^2}{2m} \sum_{i=1}^N \nabla_i^2 + \sum_{i=1}^N V(\mathbf{r}_i) + \sum_{i=1}^N \sum_{j < i}^N U(\mathbf{r}_i, \mathbf{r}_j) \right] \psi = E\psi.$$

$$\hat{H}\Psi = E\Psi$$

Each of the solution, Ψ_n has an associated eigenvalue, E_n , (real number) that satisfies the eigenvalue equation

- In this equation, m is the electron mass
- The three terms in brackets → **1) Kinetic Energy of each electron**,
 - 2) Interaction energy between each electron & the collection of atomic nuclei**
 - 3) interaction energy between different electrons**
- For the Hamiltonian we have chosen Ψ is the electronic **wave function** → is **function** of each of the **spatial coordinates** of **each** of the **N electrons** → $\Psi = \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$
- E is the **ground state energy** of the electrons
- The ground-state energy is independent of time, so this is the **time-independent Schrödinger equation**

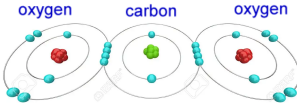
Time-dependent: $i\hbar \frac{\partial}{\partial t} \Psi = H\Psi$

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DFT Calculations



- For the Hamiltonian we have chosen Ψ is the **electronic wave function** $\rightarrow$ is function of each of the spatial coordinates of each of the **N electrons** $\rightarrow \Psi = \Psi(r_1, \dots, r_N)$
- Although the **electron wave function** is a **function** of each of the coordinates of all **N electrons**
- it is possible to approximate Ψ as a product of individual electron wave functions,
 - $\Psi = \Psi_1(r) \Psi_2(r), \dots, \Psi_N(r)$
- This expression for the wave function is known as a **Hartree product**,
- there are good motivations for approximating the full wave function into a product of individual one-electron wave functions in this fashion.

- N** $\rightarrow$ the number of **electrons**, **M** $\rightarrow$ the number of **nuclei**,
 - N >> M** $\rightarrow$ simply because each atom has one nucleus and lots of electrons.
- CO₂** **N=22, M=3**
- 
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DFT Calculations

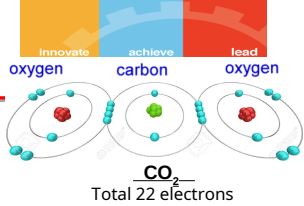


- The situation looks even worse when we look again at the **Hamiltonian, H**.
- The term in the **Hamiltonian** defining **electron-electron** interactions
- most **critical** one from the point of view of **solving** the equation.
- Means, the **individual** electron wave function we defined before, $\Psi(r)$, cannot be found without simultaneously considering the individual electron wave functions associated with all the other electrons.
- In other words, the Schrödinger equation is a **many-body problem**.
- Measurable quantity $\rightarrow$ is the probability that the N electrons are at a particular set of coordinates, $r_1, \dots, r_N$
- the quantity of physical interest is really the probability that **a set of N electrons** in any order have coordinates $r_1, \dots, r_N$

Probability
 $P = \Psi^*(r_1, \dots, r_N) \Psi(r_1, \dots, r_N)$

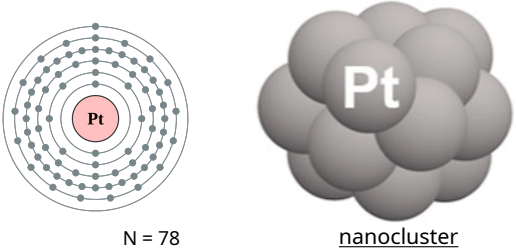
asterisk indicates a complex conjugate
The complex conjugate of a function is obtained by replacing every occurrence of $i=\sqrt{-1}$ in that function with $-i$. This procedure eliminates complex numbers in all predictions because the product is always a real number.

DFT Calculations



- N** $\rightarrow$ the number of **electrons**, **M** $\rightarrow$ the number of **nuclei**,
- If we were interested in a single molecule of **CO₂**,
- the **full wave function** is a **66-dimensional function** (3 dimensions for each of the **22** electrons)
- If we were interested in a nanocluster of 100 Pt atoms, the full wave function requires more the 23,000 dimensions!

Hurdles of solving the Schrödinger equation for practical materials

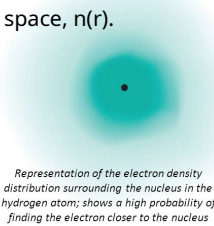


DFT Calculations



- the quantity of physical interest** is really the **probability** that **a set of N electrons** in any order have coordinates $r_1, \dots, r_N$
- A closely related quantity is the **density** of electrons at a particular position in space, $n(r)$.
- This can be written in terms of the individual electron wave functions as
- Here, the summation goes over all the individual electron wave functions that are occupied by electrons,
- so the term inside the summation is the probability that an electron in individual **wave function** $\Psi(r)$ is located at **position r**.
- The factor of **2** appears because electrons have spin and the **Pauli Exclusion Principle** states that each individual electron wave function can be occupied by two separate electrons provided they have different spins

$$n(r) = 2 \sum_i \Psi_i^*(r) \Psi_i(r)$$



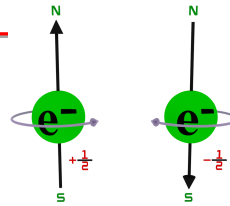


Pauli Exclusion Principle

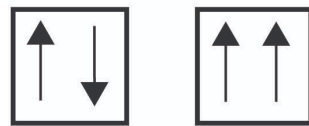
No two electrons can have the same set of quantum numbers

Two electrons in the same orbital must have opposite spins

The electron is a fermion with an electron spin of $1/2$.



Allowed Not Allowed

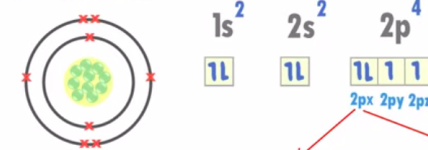


1s 1s

Purely quantum mechanical effect that has no counterpart in classical physics

no two electrons in the same atom can have identical values for all four of their quantum numbers

OXYGEN ATOM



Atomic number = 8

SET P

✓ n = 2
✓ l = 1
✓ m_l = -1
✗ m_s = +1/2

-1, 0, 1
x y z

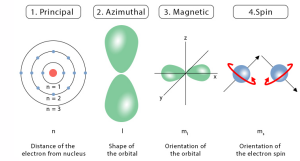
SET Q

✓ n = 2
✓ l = 1
✓ m_l = -1
✗ m_s = -1/2

-1, 0, 1
x y z

Number	Symbol	Possible Values
Principal Quantum Number	n	$1, 2, 3, 4, \dots$
Angular Momentum Quantum Number	ℓ	$0, 1, 2, 3, \dots, (n-1)$
Magnetic Quantum Number	m_ℓ	$-\ell, \dots, -1, 0, 1, \dots, \ell$
Spin Quantum Number	m_s	$+1/2, -1/2$

Quantum Numbers



DFT Calculations

$$n(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r})$$

- factor of 2 appears because electrons have spin (Pauli exclusion principle)
- This is a purely quantum mechanical effect that has no counterpart in classical physics
- the *electron density*, $n(\mathbf{r})$, which is a function of only *three coordinates*, contains a great amount of the information ...
- that is actually physically observable from the *full wave function* solution to the *Schrödinger equation*, which is a function of *3N* coordinates

DFT: From Wave Functions To Electron Density

$$\hat{H}\Psi = E\Psi \quad n(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r})$$

- The entire field of DFT rests on
 - 1) two fundamental mathematical theorems proved by **Kohn and Hohenberg**
 - 2) derivations of a set of equations by **Kohn and Sham** in the mid-1960s
- The **First Theorem**, The **ground-state energy** from Schrödinger's equation is a unique functional of the **electron density**.
- This theorem states that there exists a one-to-one mapping between the ground-state wave function and the ground-state electron density.
- To appreciate the importance of this result, you first need to know what a "*functional*" is.
- a *functional* is closely related to the more familiar concept of a function
- A functional is similar, but it takes a **function** and *defines a single number from the function*

DFT: From Wave Functions To Electron Density

- So we can restate Hohenberg and Kohn's result by saying that the ground-state energy **E** can be expressed as **$E[n(r)]$** ,
 - Where, $n(r)$ is the electron density.
- This is why this field is known as **Density Functional Theory**
- Hohenberg and Kohn Theorem**: states that the ground-state electron density uniquely determines all properties, including the energy and wave function, of the ground state.
- Why is this result important?**
- It means that we can think about solving the **Schrödinger equation** by finding a function of three spatial variables, the **electron density**, rather than a function of $3N$ variables, the **wave function**.
- Here, by "solving the Schrödinger equation" we mean, to say it more precisely, finding the **ground-state energy, E**.
- So for a nanocluster of 100 Pd atoms the theorem reduces the problem from something with more than 23,000 dimensions to a problem with just 3 dimensions.

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DFT: From Wave Functions To Electron Density

- functional described by the Hohenberg – Kohn theorem → in terms of the single-electron wave functions, **$\Psi(\mathbf{r})$** .
- The energy functional can be written as $E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$

these functions collectively define the electron density, $n(\mathbf{r})$
- The "known" terms include four contributions:

$$E_{\text{known}}[\{\psi_i\}] = -\frac{\hbar^2}{2m} \sum_i \int \psi_i^* \nabla^2 \psi_i d^3r + \int V(\mathbf{r}) n(\mathbf{r}) d^3r + \frac{e^2}{2} \int \int \frac{n(\mathbf{r}) n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' + E_{\text{ion}}.$$
- 1) the electron kinetic energies,**
- 2) the Coulomb interactions between the electrons and the nuclei,**
- 3) the Coulomb interactions between pairs of electrons, and**
- 4) the Coulomb interactions between pairs of nuclei.**

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DFT: From Wave Functions To Electron Density

- Hohenberg – Kohn theorem →
 - a functional of the electron density exists
 - that can be used to solve the Schrödinger equation,
 - the theorem says **nothing** about what the **functional** actually is.
- The **second** Hohenberg – Kohn **theorem** defines an important property of the functional:
 - The electron density that minimizes the energy of the overall functional is the true electron density corresponding to the full solution of the Schrödinger equation.**
 - If the "true" functional form were known, then we could vary the electron density until the energy from the functional is minimized, giving us a prescription for finding the relevant electron density.
- This variational principle is used in practice with approximate forms of the functional.

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DFT: From Wave Functions To Electron Density

- functional described by the Hohenberg – Kohn theorem is in terms of the single-electron wave functions, **$\Psi(\mathbf{r})$** .
- The **energy functional** can be written as $E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$

these functions collectively define the electron density, $n(\mathbf{r})$

$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

- The other term in the complete energy functional, **$E_{\text{xc}}[\{\psi_i\}]$** , is the **exchange – correlation functional**,
- it is defined to include all the **quantum mechanical effects** that are **not** included in the "known" terms.
- The Kohn – Sham equations
$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$

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$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

DFT

- The **Kohn-Sham equations**
$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$
- Comparing with the previous equation **[Full Schrödinger Equation]**
$$\left[-\frac{\hbar^2}{2m} \sum_{i=1}^N \nabla_i^2 + \sum_{i=1}^N V(\mathbf{r}_i) + \sum_{i=1}^N \sum_{j<i} U(\mathbf{r}_i, \mathbf{r}_j) \right] \psi = E\psi.$$
- the solution of the Kohn-Sham equations are single-electron wave functions that depend on only three spatial variables, $\psi_i(\mathbf{r})$
- In K-S equation, there are three potentials, V , V_H , and V_{XC} .

$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

DFT

- The Kohn-Sham equations
$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$

$$\left[-\frac{\hbar^2}{2m} \sum_{i=1}^N \nabla_i^2 + \sum_{i=1}^N V(\mathbf{r}_i) + \sum_{i=1}^N \sum_{j<i} U(\mathbf{r}_i, \mathbf{r}_j) \right] \psi = E\psi.$$
- **Full Schrödinger Equation**
- The first potential term (V) is common to both equations & in the “known” part of total energy functional
$$E_{\text{known}}[\{\psi_i\}] = -\frac{\hbar^2}{2m} \sum_i \int \psi_i^* \nabla^2 \psi_i d^3r + \int V(\mathbf{r}) n(\mathbf{r}) d^3r$$
- This potential defines the interaction between an electron and the collection of atomic nuclei
$$+ \frac{e^2}{2} \int \int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r d^3r' + E_{\text{ion}}.$$
- The second is called the Hartree potential and is defined
$$V_H(\mathbf{r}) = e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'$$
- This potential describes the Coulomb repulsion between the electron being considered in one of the Kohn-Sham equations and the total electron density defined by all electrons in the problem.

$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

DFT

- The Kohn-Sham equations
$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$
- **Full Schrödinger Equation**
$$\left[-\frac{\hbar^2}{2m} \sum_{i=1}^N \nabla_i^2 + \sum_{i=1}^N V(\mathbf{r}_i) + \sum_{i=1}^N \sum_{j<i} U(\mathbf{r}_i, \mathbf{r}_j) \right] \psi = E\psi.$$
- the Hartree potential and is defined by
$$V_H(\mathbf{r}) = e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'$$
- includes a so-called self-interaction contribution because the electron we are describing in the Kohn-Sham equation is also part of the total electron density,
- so part of V_H involves a Coulomb interaction between the electron and itself.
- The self-interaction is **unphysical**, and the **correction** for it is one of several effects that are **lumped together** into the **final potential** in the Kohn-Sham equations

$$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$$

DFT

- The Kohn-Sham equations
$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$
- V_{XC} , which defines exchange and correlation contributions to the single-electron equations.
- V_{XC} can formally be defined as a “functional derivative” of the exchange-correlation energy:
$$V_{\text{XC}}(\mathbf{r}) = \frac{\delta E_{\text{XC}}(\mathbf{r})}{\delta n(\mathbf{r})}$$
- there is something circular about our discussion of the Kohn-Sham equations

DFT

exchange and correlation contributions

innovate

achieve

lead

$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$

The Kohn –Sham equations

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r})\right]\psi_i(\mathbf{r}) = \varepsilon_i\psi_i(\mathbf{r})$$

$$V_H(\mathbf{r}) = e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'$$

there is something circular about our discussion of the Kohn –Sham equations

1) To solve the Kohn – Sham equations, we need to define the Hartree potential, and

$$n(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r})\psi_i(\mathbf{r})$$

2) to define the Hartree potential we need to know the electron density.

3) But to find the electron density, we must know the single-electron wave functions,

4) and to know these wave functions we must solve the Kohn –Sham equations.

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Kohn –Sham equations

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in the following algorithm:

1. Define an initial, trial electron density, $n(\mathbf{r})$.

2. Solve the *Kohn –Sham* equations defined using the trial electron density to find the single-particle wave functions, $\Psi(\mathbf{r})$.

3. Calculate the electron density defined by the Kohn – Sham single-particle wave functions from step 2, $n_{\text{KS}}(\mathbf{r}) = 2 \sum_i \psi_i^*(\mathbf{r})\psi_i(\mathbf{r})$

4. Compare the calculated electron density, $n_{\text{KS}}(\mathbf{r})$, with the electron density used in solving the Kohn –Sham equations, $n(\mathbf{r})$.

If the two densities are the same, then this is the ground-state electron density, and it can be used to compute the total energy.

If the two densities are different, then the trial electron density must be updated in some way.

Once this is done, the process begins again from step 2.

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exchange and correlation contributions

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$E[\{\psi_i\}] = E_{\text{known}}[\{\psi_i\}] + E_{\text{XC}}[\{\psi_i\}]$

The Kohn –Sham equations

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r})\right]\psi_i(\mathbf{r}) = \varepsilon_i\psi_i(\mathbf{r})$$

$$V_H(\mathbf{r}) = e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'$$

To break the circle → treated in an iterative way

in the following algorithm:

1. Define an initial, trial electron density, $n(\mathbf{r})$.

2. Solve the Kohn –Sham equations defined using the trial electron density

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DFT Calculations

Kohn –Sham equations

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$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}) + V_H(\mathbf{r}) + V_{\text{XC}}(\mathbf{r})\right]\psi_i(\mathbf{r}) = \varepsilon_i\psi_i(\mathbf{r})$$

How close do the two electron densities have to be before we consider them to be the same?

What is a good way to update the trial electron density?

How should we define the initial density?

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DFT Calculations



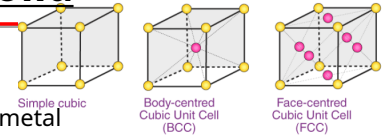
to be continued...

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DFT Calculations for Simple Solid



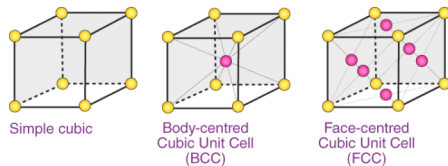
Periodic Structures, Supercells & Lattice Parameters



- Plane-wave DFT calculations
- to define the location of all atoms in a crystal of a pure metal
- Imagine, we fill 3D space with cubes of side length a , place a single metal atom at the corner of each cube. Example $\rightarrow$ Polonium
- You need value of lattice constant a
- x, y, z coordinates in 3-D space
- 1st task: We can define just *one atom* and could locate it at (0,0,0)
- 2nd task: Define a volume that fills space when repeated in all directions
- Together, these two tasks have completely defined the crystal structure of an element with the simple cubic structure
- The vectors that define the cell volume and the atom positions within the cell are collectively referred to as *supercell*

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DFT Calculations for Simple Solid



- Examples of **BCC** include **iron, chromium, tungsten, and niobium**.
- **Aluminum, calcium, nickel, copper, strontium, rhodium, palladium, silver, ytterbium, iridium, platinum, gold, lead, actinium** all have an **FCC structure**

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THANK YOU

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